Regularization and Stability: an Empirical Study of Ridge Regression

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Abstract

This report investigates whether stronger L_2 regularization increases algorithmic stability and if such stability reliably translates to improved generalization. We empirically evaluate ridge regression against Ordinary Least Squares (OLS) utilizing leave-one-out refits across three stability metrics, testing on a synthetic dataset ($d = 30$, $n = 40$) and the Diabetes dataset ($n = 50$, with a contrast at $n = 300$) over 20 repetitions. Our findings indicate that stability improves with λ up to approximately $6\times$ on synthetic data, shrinking the generalization gap in tandem (Spearman rank correlation of 0.98–0.99 over 45 dataset size and λ configurations). However, test error exhibits a U-shape concerning λ, demonstrating that highly stable models are not necessarily accurate due to underfitting. Furthermore, at a fixed λ, stability decays at a rate of approximately $1/n$, and with sufficient training data (Diabetes $n = 300$), the regularization penalty has a negligible impact on both stability and predictive performance.

1 Introduction

Classical generalization bounds in statistical learning theory largely depend on the complexity of the hypothesis class, leveraging measures such as Vapnik-Chervonenkis (VC) dimension or Rademacher complexity. An alternative framework, proposed by Bousquet and Elisseeff [1], shifts the focus from the hypothesis space to the properties of the learning algorithm itself. An algorithm is deemed stable if small perturbations to the training set—such as the removal or replacement of a single instance—do not significantly alter the resulting predictive model.

Crucially, Bousquet and Elisseeff demonstrate that a stable learning algorithm guarantees a small generalization gap. For specifically regularized algorithms, such as ridge regression, uniform stability β is theoretically bounded by $\mathcal{O}(1/(\lambda n))$, implying that higher regularization and larger sample sizes strictly limit the generalization gap. This study empirically tests this framework, guided by four primary research questions: (RQ1) How does the regularization parameter λ affect algorithmic stability in practice? (RQ2) What is the relationship between measured stability, the generalization gap, and actual test error? (RQ3) How does the size of the training set n influence stability? (RQ4) Do these theoretical phenomena hold when transitioning from controlled synthetic data to real-world datasets?

2 Background

2.1 Algorithmic stability

Let $S = \{z_1, \ldots, z_n\}$ denote a training set of size n, where $z_i = (x_i, y_i)$. We define $S^{\setminus i}$ as the modified training set formed by removing the i-th element from S. An algorithm is uniformly

β-stable if, for any training set S, any index $i \in \{1, \ldots, n\}$, and any test example z, the difference in the loss ℓ is bounded by β:

$$\sup_{S,i,z} |\ell(f_S, z) - \ell(f_{S^{\setminus i}}, z)| \leq \beta \tag{1}$$

When an algorithm is uniformly β-stable, Bousquet and Elisseeff provide a high-probability bound on the true risk R relative to the empirical risk $\hat{R}$. With probability at least $1 - \delta$:

$$R \leq \hat{R} + 2\beta + (4n\beta + M)\sqrt{\frac{\ln(1/\delta)}{2n}} \tag{2}$$

where M is an upper bound on the loss. A direct consequence of this bound is that if $\beta = \mathcal{O}(1/n)$, the generalization gap $(R - \hat{R})$ converges to zero as $n \to \infty$.

2.2 Ridge regression and why it is stable

Ridge regression applies an L_2 penalty to the model weights. Normalizing the objective by n (consistent with the formulation in [1]), the optimization problem is:

$$\hat{w} = \arg\min_w \frac{1}{n}\sum_{i=1}^{n}(y_i - w^\top x_i)^2 + \lambda\|w\|^2 = (X^\top X + n\lambda I)^{-1}X^\top y \tag{3}$$

This regularization fundamentally stabilizes the algorithm. The eigenvalues of the matrix $X^\top X + n\lambda I$ are strictly lower-bounded by $n\lambda$. Consequently, the spectral norm of its inverse is bounded by $1/(n\lambda)$.

Using the Sherman–Morrison identity, the difference between the weights trained on the full set S and the leave-one-out set $S^{\setminus i}$ can be expressed as:

$$w_S - w_{S^{\setminus i}} = (X^\top X + n\lambda I)^{-1}x_i\left(y_i - f_{S^{\setminus i}}(x_i)\right) \tag{4}$$

This equation shows that the change in weights is proportional to the leave-one-out residual multiplied by the influence of the removed point. Because the inverse matrix norm is bounded by $1/(n\lambda)$, the stability parameter β scales as $\mathcal{O}(1/(\lambda n))$. In contrast, standard Ordinary Least Squares (OLS), corresponding to $\lambda = 0$, can become highly unstable when $X^\top X$ has small eigenvalues.

3 Datasets

We utilize two datasets for our empirical evaluation. The first is a synthetic generator where features are drawn as $x \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{ij} = 0.9^{|i-j|}$ and $d = 30$. The target is $y = x^\top w^* + \varepsilon$ with standard Gaussian noise $\varepsilon \sim \mathcal{N}(0, 1)$ and a fixed true weight vector $w^* \sim \mathcal{N}(0, I)$. We generate $n_{\text{train}} = 40$ to create a challenging, data-scarce scenario, and $n_{\text{test}} = 2000$. The Bayes optimal MSE is 1.

The second is the real-world Diabetes dataset (Efron et al., 2004), consisting of 442 samples and 10 features, predicting disease progression after one year. We loaded it unscaled to prevent data leakage (as standard global pre-scaling leaks test distribution data into the training phase). For each repetition, 142 samples form the test set. We extract $n_{\text{train}} = 50$ from the remaining 300 to match the data-scarce conditions of the synthetic setup. A contrast run uses $n_{\text{train}} = 300$. The choice of $n = 50$ is deliberate: at $n = 300 \gg d = 10$, OLS is already well-conditioned and naturally stable, masking the corrective effect of regularization. The initial condition number of $X^\top X$ for a random draw highlights this: 5,470 for the synthetic data, 1,185 for Diabetes at $n = 50$, and a mere 505 for Diabetes at $n = 300$.

4 Implementation

Models and evaluation pipelines were implemented from scratch in Python using `numpy`, restricting `sklearn` solely to data loading. Ridge regression coefficients are computed via `np.linalg.solve` on the normal equations, which is computationally cheaper and more numerically stable than explicit matrix inversion. OLS is computed using `np.linalg.pinv` to ensure a minimum-norm solution when singular. The complete source code, experiment tracking, and reproducible pipelines are publicly available on GitHub at `https://github.com/PPrince33/ML-project-2026-UNiMi`.

Standardization and centring are performed strictly within the `fit` method using only the training data statistics. The unpenalized intercept is computed as the mean of the training targets ($\bar{y}$), and zero-variance features are scaled by 1 to prevent division by zero. Stability metrics are calculated using brute-force leave-one-out (LOO) refitting: n refits are performed per combination of repetition and λ.

We track three core stability metrics evaluated on the fixed test set: mean absolute prediction difference (mean $|\Delta\text{pred}|$, our primary metric for average stability), maximum absolute prediction difference (max $|\Delta\text{pred}|$, acting as a proxy for the uniform β bound), and mean absolute loss difference (mean $|\Delta\text{loss}|$, based on squared error). The codebase is verified by 41 `pytest` tests covering expected behaviors such as convergence to OLS as $\lambda \to 0$, convergence to predicting the mean as $\lambda \to \infty$, strict absence of data leakage, and exact agreement with the Sherman-Morrison LOO residual decomposition.

5 Experimental setup

We sweep λ over 30 logarithmically spaced values from 10^{-5} to 10^{2}, evaluating OLS ($\lambda = 0$) separately. All experiments are averaged over $R = 20$ repetitions. To ensure fair comparison, a single train/test split is drawn per repetition and shared across all λ values, seeded via `np.random.SeedSequence(2026).spawn`.

The empirical study consists of four primary runs:

- **Run A**: Synthetic dataset, $n = 40$.
- **Run B**: Diabetes dataset, $n = 50$.
- **Run C**: Diabetes dataset contrast, $n = 300$.
- **Run D**: Size sweep on synthetic data, evaluating $n \in \{35, 40, 50, 75, 100, 150, 200, 300, 500\}$ against $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$.

6 Results

Dataset	Model	λ	Train MSE	Test MSE	Gap	
Synthetic	OLS	0	0.244 ± 0.13	3.97 ± 1.1	3.73 ± 1.1	0
Synthetic	Ridge (best λ)	0.014	0.383 ± 0.15	2.46 ± 0.58	2.08 ± 0.61	0.1
Diabetes ($n = 50$)	OLS	0	2.1e+03 ± 4.3e+02	3.82e+03 ± 6e+02	1.73e+03 ± 8.6e+02	
Diabetes ($n = 50$)	Ridge (best λ)	0.39	2.38e+03 ± 4.1e+02	3.41e+03 ± 4.4e+02	1.03e+03 ± 7e+02	
Diabetes ($n = 300$)	OLS	0	2.82e+03 ± 1.5e+02	3.05e+03 ± 3.2e+02	233 ± 4.7e+02	0.3
Diabetes ($n = 300$)	Ridge (best λ)	0.042	2.85e+03 ± 1.5e+02	3.03e+03 ± 3.2e+02	184 ± 4.7e+02	0.3

Table 1: Performance summary comparing OLS to the optimal-parameter ridge model. Values represent mean ± std over 20 repetitions. The optimal λ is chosen a posteriori on the test set for descriptive comparison.

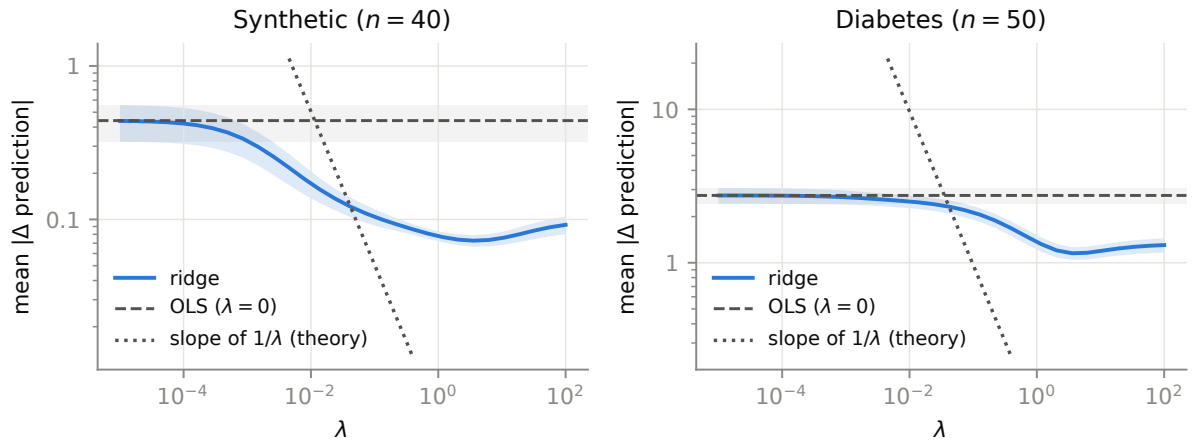


Figure 1: Mean prediction stability versus λ for the synthetic dataset ($n = 40$). Average stability improves rapidly up to $\lambda \approx 3.6$ before slightly deteriorating, forming a U-shape.

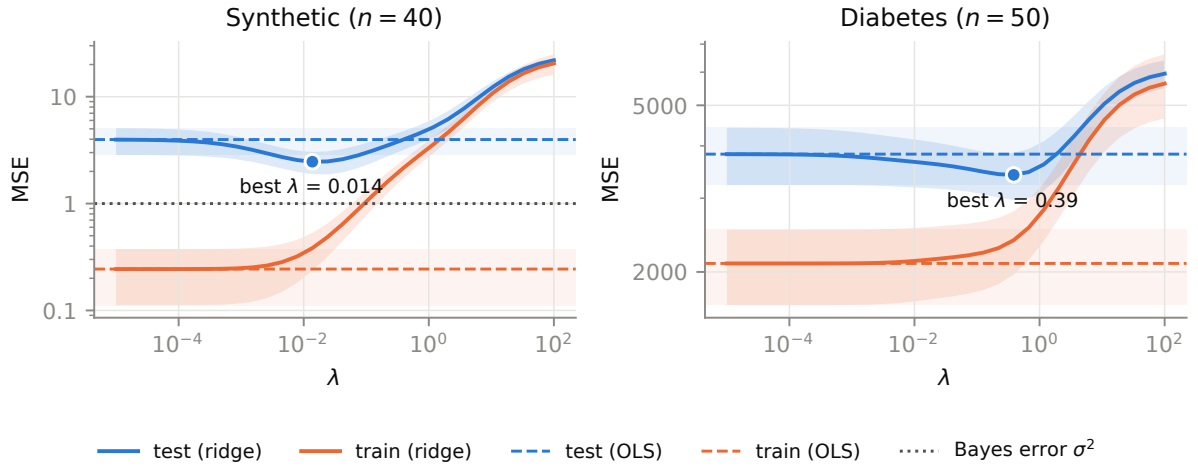


Figure 2: Test and train MSE across varying λ for synthetic ($n = 40$) and Diabetes ($n = 50$). The test error is U-shaped, balancing variance and bias, with optimal values outperforming OLS.

Figure 1 demonstrates the effect of λ on average stability. OLS ($\lambda \to 0$) yields mean $|\Delta\text{pred}|$ values of 0.441 (synthetic) and 2.75 (Diabetes). Stability sharply improves with increasing penalty, reaching a minimum at $\lambda \approx 3.6$ where values drop to 0.073 (a 6.1× improvement) and 1.15 (a 2.4× improvement). Notably, the decay slope on log-log axes is approximately -0.18 to -0.11, which is much flatter than a strict $1/\lambda$ decline.

Figure 2 plots the corresponding MSE. While training error increases monotonically as the model is constrained, test error forms a distinct U-shape. On the synthetic set, the optimal $\lambda = 0.014$ yields a test MSE of 2.46 ± 0.58 compared to OLS at 3.97 ± 1.12. For Diabetes at $n = 50$, optimal ridge achieves an MSE of 3,411 versus the OLS baseline of 3,823. Table 1 verifies that best ridge outperforms OLS in 18/20 and 15/20 repetitions for the respective datasets. However, high regularization leads to severe underfitting: at $\lambda = 100$, synthetic test MSE surges to 21.9.

Figure 3 shows the relationship between stability and the generalization gap. As λ increases to 100, the gap shrinks from 3.73 to 1.42 (synthetic) and from 1,727 to 315 (Diabetes). Across the λ sweep, stability ranks the gap exceptionally well (Spearman rank correlations of 0.94–0.97 for mean $|\Delta\text{pred}|$ and virtually 1.000 for max $|\Delta\text{pred}|$). To rule out automatic correlation due to

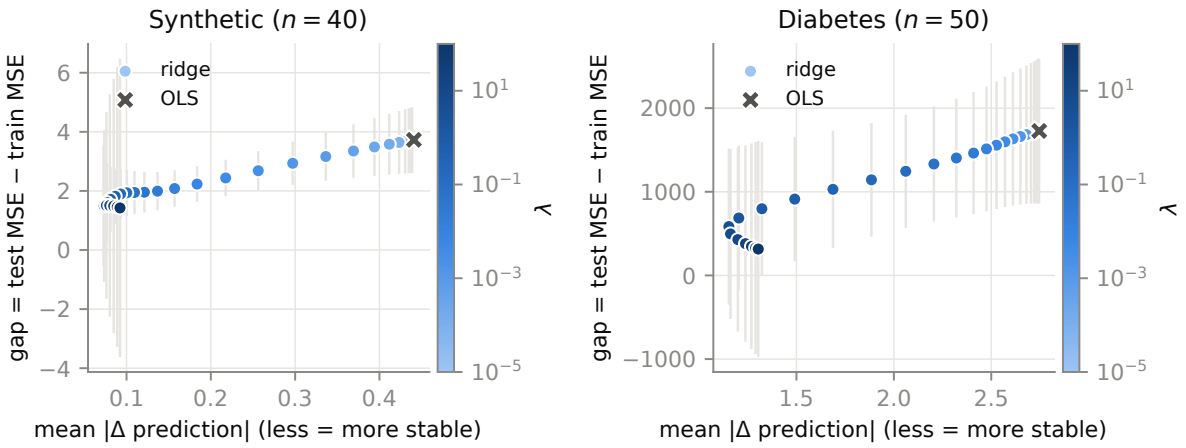


Figure 3: Correlation between generalization gap and average prediction stability for synthetic and Diabetes datasets. Both metrics decrease concurrently as λ increases.

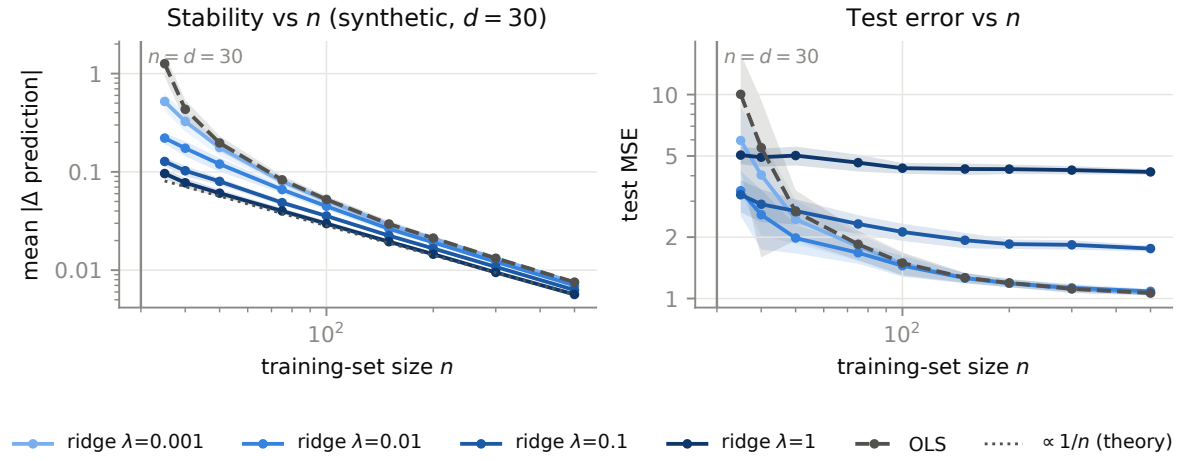


Figure 4: Average stability as a function of training set size n on the synthetic dataset, parameterized by λ. The log-log slopes confirm the theoretical $\mathcal{O}(1/n)$ decay.

monotonic scaling in a single dimension, Figure 4 varies both n and λ. Across 45 configurations, Spearman correlation remains strictly tight at 0.989. The size sweep in Figure 4 also confirms the theoretical dataset size bounds: fitted log-log slopes of stability versus n range from -1.03 to -1.38, confirming the theoretical $\mathcal{O}(1/n)$ scaling limit.

Figure 5 isolates the impact of dataset size on real-world data. Expanding the Diabetes training set from 50 to 300 naturally stabilizes OLS (2.75 down to 0.38). In this data-rich regime, the regularization penalty barely changes test error (3,054 for OLS versus 3,031 for best ridge, with ridge winning in only 11/20 repetitions). Finally, Figure 6 evaluates metric variations. While max absolute prediction difference decreases monotonically with λ (by up to 19×), mean loss stability reveals a U-shape, worsening at massive λ thresholds due to structural bias overriding variance.

7 Discussion

The experiments confirm the core predictions of Bousquet and Elisseeff's framework within practical regimes: higher L_2 regularization induces greater stability, which in turn smoothly controls the generalization gap across both synthetic and real-world datasets. The worst-case predic-

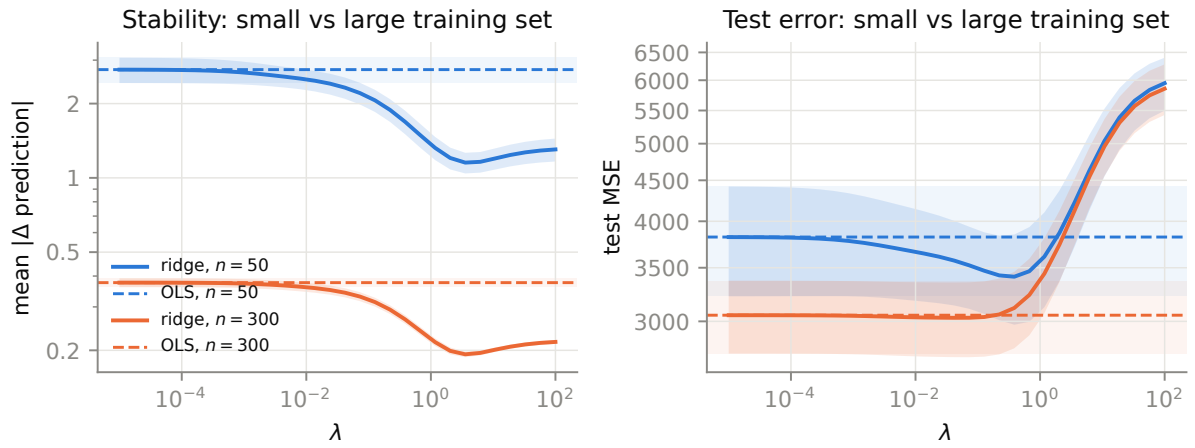


Figure 5: Comparing stability and MSE between data-scarce ($n = 50$) and data-rich ($n = 300$) scenarios on the Diabetes dataset. With ample data, OLS is naturally stable and regularisation offers minimal advantage.

tion stability, acting as an empirical proxy for uniform β, successfully decreases monotonically with λ. Furthermore, the empirical scaling with dataset size precisely mirrored the theoretical expectation of $\mathcal{O}(1/n)$.

However, the empirical evidence nuances the theory in critical ways. First, maximizing stability does not minimize test error. Massive regularization ($\lambda = 100$) guarantees a highly stable, small-gap model, but at the cost of catastrophic underfitting (test MSE degrades 9× worse than Bayes optimal). The optimal λ resides at the lowest point of the test error U-shape, balancing stability gains against induced bias. Second, average stability decays at a significantly slower rate than $1/\lambda$. This occurs because a specific feature direction only enters the $1/(n\lambda)$ variance-reduction regime once $n\lambda$ exceeds the associated eigenvalue of $X^\top X$, which can be several hundreds in ill-conditioned, data-scarce setups.

Finally, the metrics indicate bounded limits to the theory's assumptions. Mean prediction stability eventually forms a U-shape plateau because the intercept is left unpenalized; at infinite λ, predictions collapse to the leave-one-out mean $\bar{y}_{\setminus i}$, leaving a residual instability floor strictly dictated by $1/n$. Furthermore, with abundant, well-conditioned data ($n = 300 \gg d$), the inherent geometric stability of OLS leaves little room for ridge regression to significantly improve predictive performance.

8 Conclusion

Algorithmic stability provides a rigorous theoretical lens for why regularization aids generalization. Stronger penalties restrict hypothesis variance, forcing algorithms to remain stable across training set perturbations, which strictly bounds the discrepancy between empirical and true risk. However, stability alone is insufficient for model selection: while infinite regularization yields perfect stability, it destroys predictive utility.

Practically, L_2 regularization is most impactful in data-scarce or ill-conditioned regimes where base algorithms like OLS suffer from wild variations. Future work should explore whether these dynamics hold for non-differentiable L_1 penalties (Lasso), analyze behavior under varying feature dimensionalities d, and extend evaluations from leave-one-out to broader replace-k stability paradigms.

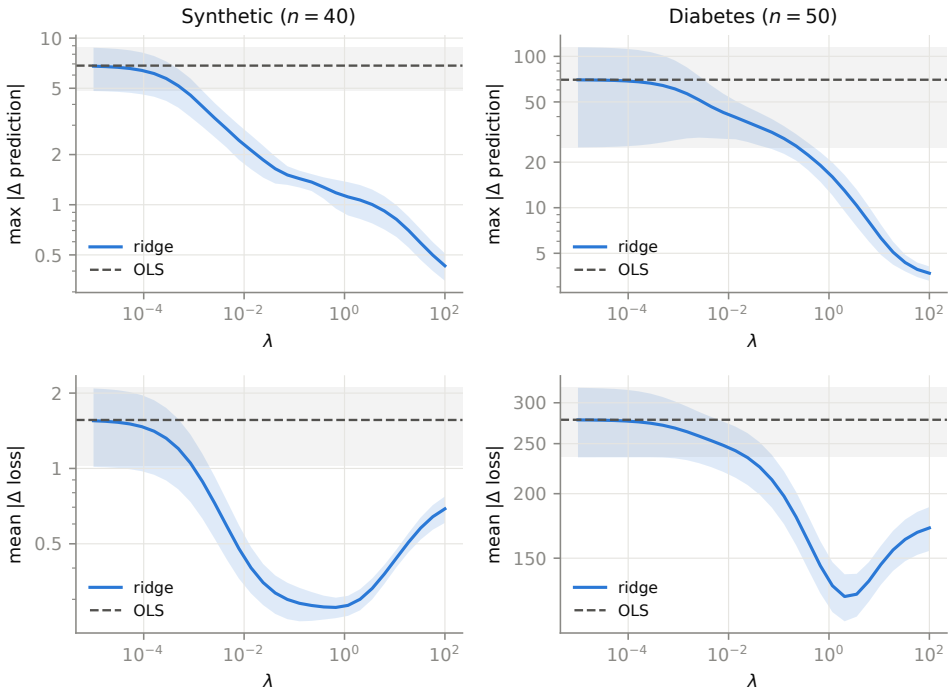


Figure 6: Alternative stability metrics vs λ. The worst-case prediction stability (proxy for uniform β) decreases monotonically, while mean loss stability follows a U-shape.

References

[1] Olivier Bousquet and André Elisseeff. Stability and generalization. *Journal of Machine Learning Research*, 2(Mar):499–526, 2002.

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